Credit Name: CSE2130 - File Structures & Exception Handling
Assignment Name: Student Semester GUI

Error Log Entry

1. What error message did you encounter (if any)?

I did not encounter a specific error message while running the program, but entering letters or leaving a grade field empty can cause a NumberFormatException.

2. What unexpected behaviour did your program exhibit?

If a user enters letters instead of numbers for a grade, the program cannot calculate the average. Without error handling, this could cause the program to stop executing the button's action.

3. What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

The issue is caused by Double.parseDouble(), which converts the text entered into a number. If the text is not a valid number, Java throws a NumberFormatException. I handled this by using a try-catch statement.

4. Include a screenshot of specific lines of code.

```
double grade1 = Double.parseDouble(grd1.getText());
double grade2 = Double.parseDouble(grd2.getText());
double grade3 = Double.parseDouble(grd3.getText());
double grade4 = Double.parseDouble(grd4.getText());

double average = (grade1 + grade2 + grade3 + grade4) / 4.0;


} catch (NumberFormatException ex) {
                        JOptionPane.showMessageDialog(
                           frame,
                           "Please enter a valid number for each grade."
                        );

                    }
```

5. How did you fix the issue?

I used a try-catch statement to catch the NumberFormatException. If the user enters invalid grade information, the program displays a message asking them to enter valid numbers instead of allowing the error to go unhandled.

6. Provide the corrected code or solution using a screenshot.

```
try {
   double grade1 = Double.parseDouble(grd1.getText());
```

```
    double grade2 = Double.parseDouble(grd2.getText());
    double grade3 = Double.parseDouble(grd3.getText());
    double grade4 = Double.parseDouble(grd4.getText());

    double average = (grade1 + grade2 + grade3 + grade4) / 4.0;

    percentAvg.setText(String.format("%.1f%%", average));

} catch (NumberFormatException ex) {
    JOptionPane.showMessageDialog(
        frame,
        "Please enter a valid number for each grade."
    );
}
```